Khai-Nguyen Nguyen

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Education

University of Virginia — Virginia, USA
Ph.D. Student in Data Science — 2026 - 2029 (expected)
- **Advisor:** Dr. Chirag Agarwal

College of William and Mary — Virginia, USA
MSc Student in Computer Science — 2023 - 2026
- **GPA:** 3.96/4.00
- **Advisor:** Dr. Antonio Mastropaolo

Bucknell University — Pennsylvania, USA
B.Sc. in Computer Science and Engineering, minor in Statistics — 2019 - 2023
- **GPA:** 3.94/4.00 *(Summa Cum Laude)*

Research Interests

AI Safety; Evaluation & Interpretability of LLMs and Multimodal LLMs; Robustness of AI Systems in High-Stakes Domains (e.g., Healthcare, Software Engineering)

Selected Skills

Programming Languages: Python, Java, C/C++, R, SQL
Libraries and Frameworks: PyTorch, TensorFlow, Pandas, HuggingFace, vLLM, nnsight, TransformerLens, SentenceTransformers, DSPy, Matplotlib, Seaborn, GurobiPy

Selected Publications

♣: *equal contribution*

1. **Vision Language Models are Biased** [pdf]
   An Vo♣, **Khai-Nguyen Nguyen♣**, Mohammad Reza Taesiri, Vy Tuong Dang, Anh Totti Nguyen, Daeyoung Kim
   ICLR 2026
2. **Pattern over Pixels: Measuring Pattern Completion Bias in Multimodal Code Generation** [pdf]
   Khai-Nguyen Nguyen, Oscar Chaparro, Antonio Mastropaolo
   ASE 2026
3. **Toward Explaining Large Language Models in Software Engineering Tasks** [pdf]
   Antonio Vitale♣, **Khai-Nguyen Nguyen♣**, Denys Poshyvanyk, Rocco Oliveto, Simone Scalabrino, Antonio Mastropaolo
   ACM Transactions on Software Engineering and Methodology (TOSEM), 2026
4. **Sentiment Reasoning for Healthcare** [pdf]
   Khai-Nguyen Nguyen♣, Khai Le-Duc♣, Bach Phan Tat, Duy Le, Truong-Son Hy
   ACL 2025, Industry Track (Oral)
5. **Medical Spoken Named Entity Recognition** [pdf]
   Khai Le-Duc, David Thulke, Hung-Phong Tran, Long Vo-Dang, **Khai-Nguyen Nguyen**, Truong-Son Hy, Ralf Schluter
   NAACL 2025, Industry Track
6. **Real-time Speech Summarization for Medical Conversations** [pdf]
   Khai Le-Duc♣, **Khai-Nguyen Nguyen♣**, Long Vo-Dang, Truong-Son Hy
   Interspeech 2024 (Oral)
7. **Getting away with network pruning: From sparsity to geometry and linear regions** [pdf]
   Jeffrey Cai♣, **Khai-Nguyen Nguyen♣**, Nishant Shrestha, Aidan Good, Ruisen Tu, Xin Yu, Shandian Zhe, Thiago Serra
   CPAIOR 2023
   Sparsity in Neural Networks Workshop @ICLR 2023

Preprints

♣: equal contribution

1. **Prior Bias in Vision Language Models on UML Diagram Interpretation** [pdf]
   Zaiyu Cheng♣, **Khai-Nguyen Nguyen♣**, Antonio Mastropaolo
   arXiv 2026
2. **S-Chain: Structured Visual Chain-of-Thought for Medicine** [pdf]
   Khai Le-Duc, Phuong T.H. Trinh, Duy Minh Ho Nguyen, Tien-Phat Nguyen, Nghiem Tuong Diep, An Ngo, Tung Vu, Trinh Vuong, Anh-Tien Nguyen, Nguyen Dinh Mau, Van Trung Hoang, **Khai-Nguyen Nguyen**, Hy Nguyen, Chris Ngo, Anji Liu, Nhat Ho, Anne-Christin Hauschild, Khanh Xuan Nguyen, Thanh Nguyen-Tang, Pengtao Xie, Daniel Sonntag, James Zou, Mathias Niepert, Anh Totti Nguyen
   arXiv 2025

Research Experiences

AURA Lab, College of William and Mary, Virginia, USA

Research Assistant. Advisor: Prof. Antonio Mastropaolo 2024.10 - present

Paper: *Pattern over Pixels: Measuring Pattern Completion Bias in Multimodal Code Generation*

- Built `Pattern2Code`, the first benchmark measuring pattern-completion bias in screenshot-to-code generation, by perturbing repeated UI structures in real webpages
- Evaluated five frontier MLLMs, showing they systematically override visual evidence to complete familiar UI patterns, and analyzed reasoning traces to characterize failure modes

Paper: *Prior Bias in Vision Language Models on UML Diagram Interpretation*

- Co-led a counterfactual benchmark reversing UML relation arrows, showing open-source VLMs lose 33% accuracy on average and rely on pretrained priors over visual evidence

Paper: *Toward Explaining Large Language Models in Software Engineering Tasks*

- Implemented a model-agnostic SHAP-based framework to quantify the effect of each input feature of software engineering prompts on the model output
- Proposed and conducted quantitative experiments to understand the attribution capabilities of our framework compared to random and LLM baselines

Paper: *Resource-Efficient & Effective Code Summarization*

- Built a PEFT training pipeline for code language models (e.g., `DeepSeek-Coder`, `CodeLlama`) to quantify the impact of QLoRA on code summarization

Nguyen Lab, Auburn University, Alabama, USA

Remote Collaborator. Advisor: Prof. Anh Totti Nguyen 2025.01 - present

Paper: *Vision Language Models are Biased*

- Conducting ablation and activation patching experiments on open-source VLMs using the `nnsight` package to locate the source of bias in the model
- Built an AI-assisted image generation pipeline for realistic-looking counterfactual images (e.g., animals, logos and flags) using `Gemini-2.0-Flash`
- Conducted ablation experiments (e.g., linear probing, attention visualization, image background removal with `LangSAM`) to locate contributing factors to knowledge bias

FPT Software AI Center, Vietnam
Remote Research Intern. Supervisor: Prof. Truong-Son Hy 2024.01 - 2024.09

Paper: *Sentiment Reasoning for Healthcare*

- Proposed the *sentiment reasoning* task and implemented a chain-of-thought finetuning pipeline for LLMs in medical sentiment analysis
- Co-led the construction of the `Sentiment Reasoning` dataset consisting of 30,000 samples across 5 different languages

Paper: *Real-time Speech Summarization for Medical Conversations*

- Co-led the construction of the `VietMed-Sum` dataset for medical speech summarization consisting of 24,000 samples and built its training pipeline
- Proposed and conducted an ablation study on the effect of mixing human-curated and synthetic data for medical summarization in budget-constrained scenarios

Analytics Lab, Bucknell University, Pennsylvania, USA
Undergraduate Research Assistant. Advisor: Prof. Thiago Serra 2022.05 - 2023.01

Paper: *Getting away with network pruning: From sparsity to geometry and linear regions*

- Co-developed a theorem establishing an upper bound on the expressiveness, quantified by linear regions, of ReLU-based deep neural networks.
- Implemented `GurobiPy`-based functions that identify stably active and inactive neurons, and count the linear regions in deep neural networks

Industry Experiences

Machine Learning Intern, *CodaMetrix* | Boston, USA 2025.05 - 2025.08

- Developed a Gemini-based judge to evaluate and refine medical entity annotations, improving performance by 2–4% in F1 over human baseline
- Designed and deployed entity extraction LLMs with DSPy prompt optimization on Databricks, achieving an overall F1 of 70% on `Llama-3.1-70B`

Machine Learning Intern, *CodaMetrix* | Boston, USA 2024.05 - 2024.08

- Trained a BERT-based LLM for ICD-10 multilabel classification, outperforming the deployed system by 4% in accuracy
- Accelerate training by 4x on Databricks using distributed and parallel training

Impact & Press coverage

Benchmark adoption: `VLMsAreBiased` used in official model evaluations by Google DeepMind (Gemini 3 Pro) and ByteDance (Seed 1.8)

SkyNews: AI could be giving US lethal edge in Iran war - but there are dangers

SkyNews: This “chicken” test shows why AI could be so deadly in war